



The 10-Minute Walkthrough

Eight features and the novelty, in the order to show them.

Start here

- OceanEmbed reconstructs ocean **temperature at 15 depths, from the surface down to 1000 m**, over the North Indian Ocean, using **only what a satellite can see**.
- Everything below runs in **one page**. Nothing to install between features, no second tab to open.

RUN THIS ONCE, BEFORE THE JURY WALKS IN

```
cd oceanembed
.venv\Scripts\python.exe -m streamlit run app/ui/main.py --server.port 8500
```

- It opens at **http://localhost:8500**. Allow about a minute on the first feature -- the reconstruction is cached per date, so every feature after it is instant.
- Leave **Input** on **Satellite**. The GLORYS setting is a comparator that reads reanalysis, which the problem statement does not allow. The app warns you if it is on.

WHAT IS ON SCREEN

A **top bar** -- date, input, GPU, and a **Summary** button -- and then three rows of buttons: **SEE IT**, **PROVE IT**, **STRESS IT**. Clicking a button swaps what is on the stage. It never reloads the page, so you can move back and forth freely while answering a question. Every control has a **?** next to it holding the caveat for that number.

The eight, in order

- **SEE IT** shows what the system produces, **PROVE IT** shows it is right, **STRESS IT** shows what happens when you attack it. Walk them in that order and the jury never has to ask 'but is it real?' -- you get there first.

1. 3-D Ocean	
Click	SEE IT → 3-D Ocean
Shows	The whole basin as a solid volume you can spin, with a cyan sheet floating inside it. That sheet is the 26 °C layer -- the depth of warm water a cyclone burns.
Say	"This is one day of ocean, reconstructed from satellite alone. Nothing here was measured below the surface."
Careful	It is a reconstruction, not an observation. Say so first, before anyone asks.

2. Click a point	
Click	SEE IT → Click a point
Shows	Click anywhere on the sea and the full temperature profile under that point appears -- all 15 depths, surface to 1000 m.
Say	"Any pixel in the basin, any day, all the way down. This is the product."
Careful	Click on land or outside 5–30°N, 45–105°E and it refuses rather than guessing. That refusal is deliberate -- show it if you have a spare ten seconds.

3. Confidence	
Click	SEE IT → Confidence
Shows	The same field, but colour is temperature and fade is doubt . Vivid where the model is sure, washed out where it is not.
Say	"Every value ships with an uncertainty band. We report its coverage as a range, 80.1% to 96.0% by depth , not as a single average, because the average would hide the worst depth."
Careful	Do not call it calibrated. It is improved, not calibrated -- at 50 m the band covers 80% where a perfect band would cover 95%.

4. Validation	
Click	PROVE IT → Validation
Shows	The model's answers next to 962 real Argo float profiles it never trained on , depth by depth.
Say	" 0.9006 °C average error against 962 independent floats, over 12,736 depth comparisons. It beats the seasonal average at 14 of 15 depths ."
Careful	At 1000 m the seasonal average wins by 0.055 °C. Volunteer it -- the chart labels it anyway, and a jury that finds it themselves stops believing the other fourteen.

5. Cyclone heat	
Click	PROVE IT → Cyclone heat
Shows	Not surface warmth but the heat stored in the column -- the quantity that decides whether a storm intensifies.
Say	"A satellite sees a warm surface. It cannot see whether the warmth is two metres deep or two hundred. That difference is what this gives you, and it is what forecasters actually need."
Careful	The argument for the whole project. If you show only one PROVE IT feature after Validation, make it this one.

6. Where to measure next	
Click	PROVE IT → Where to measure next
Shows	A map of where another float or buoy would teach the model the most: high model doubt meeting an energetic ocean .
Say	"The system does not just answer questions. It tells you where its own answer is weakest, so the next measurement is not wasted."
Careful	Call it a suggestion, not an instruction . It is a heuristic, and the app says so.

7. Cloud cover	
Click	STRESS IT → Cloud cover
Shows	Infrared cannot see through monsoon cloud, so this blanks part of the input on purpose and measures what the loss costs.
Say	"We hid 15% of the sea-surface temperature and the score did not fall -- 0.8950 against 0.9078 . The model leans on sea-surface height and salinity too, so cloud does not blind it."
Careful	Those two figures are from an older scoring protocol than the 0.9006 headline. Compare them with each other, never with the headline.

8. Cyclone wake	
Click	STRESS IT → Cyclone wake
Shows	A real storm's track laid over the reconstructed heat, before and after it passed.
Say	"Cyclone Shakhti, October 2025. The ocean cooled at 11 of 12 track points , on average 4.26 kJ/cm² . Nobody taught the model that storms cool the ocean -- it was never given a storm. It reproduced the wake from satellite input alone."
Careful	This is the strongest single moment in the demo. Slow down here.

The novelty, in one page

- Reconstructing subsurface temperature from the surface is **not new**, and we say so in writing. These three are ours, and all run on the **frozen shipped model**.

	What it is	The number
Physical profiles STRESS IT → Physical profiles	Water must get denser with depth, or the column would turn over. Everyone else adds a penalty that makes breaking it <i>rare</i> . We project the answer so breaking it is <i>impossible</i> , then recompute the density to check.	805 violations → 0 , verified by recomputing rather than asserted. Costs 0.0002 °C.
What it can see STRESS IT → What it can see	We differentiate the model to ask which surface signal it actually reads at each depth. It uses sea-surface temperature for the shallow layer and sea-surface height for the thermocline -- which nobody taught it .	Signal reaches a median depth of 1000 m ; the weakest quarter of profiles fade by 500 m .
Learn from a float PROVE IT → Learn from a float	A float surfaces with a real measurement. We correct the model's internal state and the fix travels to water in a similar state -- not to water that merely happens to be nearby. No retraining .	Error falls 0.0150 °C at similar states and rises 0.2477 at dissimilar ones -- which is the control that proves it is the similarity doing the work.

IF A JUROR ASKS WHAT IS GENUINELY NEW

"The reconstruction itself is published prior art and we cite it. What is ours is a **guarantee** instead of a penalty, a **measurement of what the model can and cannot see**, and a way to **use a new observation without retraining**. Our claim is system-level, and it is written down that way."

How to finish

- 1 Press **Summary** in the top bar. It opens the one-screen account of what was built and what it scores -- read from the frozen artifact, not typed into a slide.
- 2 Close on the honest limit, before they ask for it: **"the model runs slightly warm, the thermocline is our hardest depth, and one depth of fifteen is beaten by the seasonal average. All three are on the charts."**
- 3 Then the one sentence to leave in the room.

It reconstructs the ocean underneath a satellite image, it tells you how much to trust each value, and it tells you where to measure next.

IN ONE LINE

If you remember nothing else: one command, three bands, eight clicks.

8500

THE ONLY PORT

0.9006 °C

AGAINST 962 FLOATS IT NEVER SAW

14 of 15

DEPTHS BEATEN

Detail for any single feature lives in its own note, 02 to 18. The limitations and the viva answers are in note 19.